

BONES VIBRATION

AI Music Creator

Prompt Control Room

User guide • Version 0.9.0

Dense Suno-oriented prompts, local track and image analyzers, EBU R128 loudness metering, and studio WAV export. Runs in the browser or as a Windows desktop app.

AI Music Creator - Prompt Control Room

Version 0.9.0

A Next.js app for building dense, reproducible prompts for AI music workflows (especially Suno-like layouts): genres, grooves, sounds, lyric direction, presets, optional reference analyzers, and export blocks that respect Style / Lyrics field limits. Ships as a static web app and an optional Electron Windows installer.

Highlights (v0.9.0)

- Architecture refactor - useReducer project state, persistence/actions hooks, three-column workspace layout, and co-producer engine extracted from the main page (~780 lines vs ~2,035).
- Reliability - route error boundary, 60s Co-Producer LLM abort/timeout, safeLocalStorage, pinned deps, release CI runs full npm run check.
- Performance - memoized workspace context, React.memo on heavy panels, prompt pipeline keyed on analyzer summary only (not waveform peak edits).
- Tests - 101 unit tests (audio cache + IndexedDB, co-producer engine, audio analyzer) plus Playwright e2e smoke coverage.

Highlights (v0.8.4)

- Reset clears Lyric Style Prompt - blank placeholder until vocal mode and theme are set; generated lyrics/hooks cleared on reset.
- E2e + unit tests for lyric prompt blank slate after Reset to Default.

Highlights (v0.8.3)

- Analyzer catalog sync - audio/image DNA suggestions map to Suno genre/instrument/rhythm pills (BPM-aware).
- 8 new factory presets - Trap, Pop, Lo-Fi, Reggaeton, Metal, Jazz, Afrobeats, Detroit Techno.
- Expanded vocal roles - whispered, raspy, autotuned, duet, crowd chant, stacked harmonies.
- Co-Producer LLM - Spanish/non-English section-tag rules in system prompts.
- Lyric prompt cleanup - language rules without redundant boilerplate.

Highlights (v0.8.2)

- Suno lyric languages - full language catalog (strong + extended tiers), grouped Language selector, Suno section-tag rules in lyric prompts and Co-Producer.
- Suno genres & instruments - ~130 genres, ~150 instruments/textures, searchable grouped pickers; Style Prompt Library still adds 900+ genre-wheel fusion phrases.
- CI e2e - Playwright smoke tests run on every push/PR alongside unit tests and build.

Highlights (v0.8.1)

- Reset to Default !' blank slate - clears all preselected genres, sounds, rules, and lyrics so you can build step by step from guided step 1 (factory presets stay in the sidebar).
- Reset e2e test - Playwright coverage for blank-slate reset behavior.

Highlights (v0.8.0)

- Co-Producer Generate Lyrics - after picking Lyric Style, generate editable lyric drafts matched to that style prompt (Another take, stale-style warning, char budget).

- Style-aware hooks - hook sketches tied to the same Lyric Style presets.
- Instrumental !' lyrics - timed scaffold plus Co-Producer singable draft and hooks in one step.
- Optional LLM backend - OpenAI-compatible API (key stored locally); falls back to built-in templates.
- Swedish / mixed language phrase pools per style.

Highlights (v0.7.9)

- Studio export fix - Electron desktop no longer stalls at 5%; falls back to main-thread export when Web Workers are unavailable on file://.

Highlights (v0.7.8)

- Lyrics on instrumentals - after uploading a track, Add lyrics timed to this track builds a [Verse]/[Chorus] scaffold locked to BPM, duration, and highlight/drop.
- Auto vocal switch - leaves Instrumental mode, merges track DNA, strips “no vocals” rules, and jumps to the Lyric direction step.

Highlights (v0.7.7)

- Cleanup tooling - npm run cleanup:dist:sudo and npm run cleanup:sudo for locked electron-dist* folders on Windows.
- Version fallback sync - in-app version matches package.json when env injection is unavailable.
- Export tests - WAV 24-bit filename and PCM header coverage in vitest.

Highlights (v0.7.6)

- WAV 24-bit export - honest third studio format alongside 16-bit WAV and MP3.
- Undo keeps compact waveforms - peak arrays "d4096 samples restore on revert.
- Desktop update controls - “Check for updates” and “Restart to install” in the packaged app header.
- Pinned Electron toolchain - electron@41.3.0 and electron-builder@26.8.1 for reproducible builds.

Highlights (v0.7.5)

- Undo snapshot fix - capture uses the correct slim state (guided step, variations, history).
- Honest studio export - WAV and MP3 only; legacy flac requests map to WAV.
- Release workflow - push a v* tag to publish the Windows installer and latest.yml for auto-update.
- Vitest 4 - clears the npm audit advisory on the test runner.

Highlights (v0.7.4)

- Turbopack root - turbopack.root in next.config.js fixes dev crashes when Next mis-detects app/ as the workspace.
- Dev debug - npm run debug uses a single inspector on port 9241 (scripts/debug-dev.ps1); npm run stop also clears stray Next node processes.
- Undo - Snapshot restores guided step, variations, and prompt history; audio rehydrates from IndexedDB when cached.
- MP3 fallback - If MP3 encode fails in the worker, export falls back to WAV with a status message.
- npm run cleanup:dist - Removes locked electron-dist* folders when Windows/Cursor releases them.

Highlights (v0.7.2)

- Analyzer honesty - In-app banners clarify that track/image scans are local heuristics, not ML classification.
- Lyrics priority trim - Suno Lyrics paste matches Style: theme/language/sections first, long bodies trim from the end ("d5000).

- Studio export - Background worker with progress bar; WAV / MP3; highlight-loop export.
- Undo snapshot - Revert to last snapshot before preset load, import, merge, or variations.
- Variation A/B - Side-by-side compare with changed-line summary.
- CI - GitHub Actions runs npm test + npm run check on push/PR.
- Electron auto-update - Checks GitHub releases when running a packaged build.

Highlights (v0.7.1)

- Guided Suno path - Step-through workflow, Polish step, progressive style preview, Style capped at 1000 characters with priority ordering on copy.
- Expanded English style vocabulary - Large curated catalog including world/regional styles paired with instruments, sound-design FX, environment beds, orchestral and band instruments, moods, and fusion labels; English-only picker with dedupe.
- Track analyzer (local) - Drop WAV / MP3 / OGG / M4A for a Sonoteller-style report: editable summary, genres, moods, instruments, BPM/key estimates, and a highlight section with full-track + zoomed waveforms (drag amber handles to set the range).
- Audio DNA !' Suno - Merge into Suno fields applies tempo, genres, sounds, rhythms, mood sliders, a compact AUDIO: rule line, and copies the track summary into Goal and Notes when present.
- Image analyzer - Drop JPG / PNG for palette-driven genre/sound suggestions; merges into IMAGE: rule lines and guided fields.
- LUFS meter (EBU R128) - After attach/analyze, shows gated integrated LUFS and true peak (dBTP) using a BS.1770-4-style engine (libebur128-aligned K-weighting and oversampling).
- Studio WAV export - Three local mastering presets: Streaming (" 14 LUFS integrated + " 1 dBTP limit), Wide spatial (stereo width), Punch (low-end and dynamics). Exports 16-bit stereo WAV in the browser.
- Waveform persistence - Autosave/history omit heavy peak arrays; IndexedDB caches audio for rehydrate. Export JSON keeps full waveformPeaks; import preserves them. Attach audio reconnects playback when the cache is missing.
- Version-aware reset - A major package.json version bump clears the autosaved project and analyzer state; presets and history are kept. Patch/minor bumps migrate the saved project in place.
- Refactored UI - Analyzer logic in use-analyzers, splash/header in app-shell, splash timing via useSyncExternalStore (no hydration mismatch in dev).
- Packaged assets - public/bones-logo.webp, root icon.ico, and build/AI_Music_Creator_README.pdf included for Electron builds.
- Live length readout - Style box and lyrics direction character counts next to analyzers (same strings as the validator).
- Presets & history - Factory and custom presets, project save/import/export JSON, variation engine, Co Producer helpers, Suno language index and symbol guides.

Requirements

- Node.js 18+ recommended
- npm (ships with Node)

Development

```
npm install
npm run dev
```

Open <http://localhost:3000>.

Debug (Next.js + Node inspector)

```
npm run debug
```

Starts next dev --inspect=9241 (one inspector port). In VS Code / Cursor, attach to 9241 (see .vscode/launch.json).

Stop stuck dev/debug ports on Windows:

```
npm run stop
```

Remove stale packaging folders (when not locked):

```
npm run cleanup:dist
```

If folders stay locked (Cursor/Electron holding files), run elevated cleanup - it removes what it can and schedules the rest for delete on next reboot:

```
npm run cleanup:dist:admin
```

Old installer output under dist/ is gitignored; safe to delete locally anytime.

Electron auto-update

Packaged builds check GitHub Releases for Druttzen/ai-music-tool on startup. Updates require a published release with latest.yml from electron-builder --publish (or manual upload). Dev/npm run electron skips update checks.

Production build

```
npm run build
```

Pre-ship check (ESLint, zero warnings, then production build):

```
npm run check
```

npm run lint runs ESLint alone.

Static export output is written to out/ (see next.config.js - assetPrefix: "/" for Electron-friendly relative assets).

Desktop (Electron)

```
npm run dist
```

Runs npm run build, regenerates build/AI_Music_Creator_README.pdf from this README (npm run build:readme-pdf), prepares out/ for Electron (npm run prepare:electron-dist), then electron-builder. Installer output under electron-dist/ (see package.json build section). The PDF opens once on first launch (main.js).

Regenerate the PDF alone:

```
npm run build:readme-pdf
```

If electron-dist/win-unpacked is locked, prepare:electron-dist automatically falls back to electron-dist-fresh, electron-dist-v{version} (e.g. electron-dist-v071), or a timestamped electron-dist-build-* folder. Close any running AI Music Creator instance and Explorer windows in old output folders if you want the default electron-dist/ path back.

Saved data

Storage | What

localStorage | Autosaved project (ai_music_creator_visual_tool_v3), custom presets, prompt history (slim snapshots - no waveform peaks in history/autosave)

sessionStorage | Splash intro seen flag

IndexedDB (ai-music-creator) | Cached audio blobs for waveform rehydrate (not in exported JSON)

Export JSON | Full project including audioAnalysis.waveformPeaks; re-import restores peaks and timeline (attach audio on another machine for playback if cache is empty)

Studio export notes

- Processing runs locally in the browser; long tracks can take time (max 10 minutes per export).
- Streaming targets " 14 LUFS integrated loudness - common for Spotify/YouTube-style delivery, not a substitute for a certified broadcast meter or mastering engineer.
- Wide spatial is stereo enhancement, not Dolby Atmos.

Version source of truth

The UI reads APP_VERSION from package.json via NEXT_PUBLIC_APP_VERSION in next.config.js. Bump package.json version for releases; the fallback in app/lib/music-config.js should match for dev without env.

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